----- Active space -----

	Bacterial vaginosis	Candidiasis	Trichomonas vaginitis
DOC : pregnancy	• metronidazole (DOC) : – Ist trimester : Vaginal metronidazole – 2nd/3rd trimester : Oral metronidazole	• Oral fluconazole : Contraindicated • Topical clotrimazole : Doc	• metronidazole – Ist trimester: Vaginal metronidazole – 2nd/3rd trimester: Oral metronidazole – Dose : Same
Partner Rx	X	For symptomatic partners only	✓

• Spores of candida : Smaller than nucleus.

Cervicitis

00:24:07

Ectocervix vs Endocervix :

	Ectocervix	Endocervix
Lined by	Stratified squamous epithelium (Like vagina)	Columnar epithelium
Infections	All vaginal infections may be present : • Candida • Trichomonas • HSV • HPV	• Chlamydia • Gonorrhoea (Infection of endocervix → Cervicitis)

CHLAMYDIAL & GONOCOCCAL CERVICITIS

m/c route of spread : Ascending infection along with sperms (STD).
m/c site : Endocervix > Urethra

Clinical presentation :

- Asymptomatic (50–80%).
- Discharge (m/c presentation) :
  - Mucoid/purulent.
- Dysuria.
- Dyspareunia.
- Post coital bleeding.

----- Active space -----

Per speculum examination :

- Unhealthy, friable cervix.
- Bleeds on touch.
- Erosion/ulcer.

Per vaginal examination :

- Uterus : Normal.
- Adnexa : Normal.

D/d : PID

- Uterine tenderness.
- Adnexal tenderness.

	Chlamydia	Gonorrhoea
Gram stain	• >10 Neutrophils/field • No organism	• Gram -ve diplococci • Intra/Extracellular
IOC	Nucleic acid amplification test	
Specimen	• Endocervical swab • Urine if urethritis +ve	
Gold standard	Culture on McCoy cell lines	Culture on Thayer Martin media
DOC : Non-pregnant	Tab. Doxycycline 100mg OD x 7 days	Inj Ceftriaxone 150mg IM single dose
Pregnant	• Doxycycline : C/I • Tab. Azithromycin 1g oral : Single dose	

Complications :

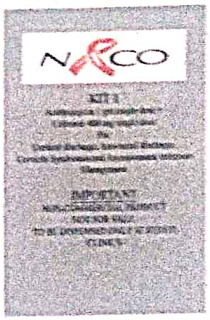
- PID : D/t ascend of infection.
- Perihepatitis : Fitz Hugh Curtis syndrome.

MANAGEMENT

Syndromic management :

- Advised by NACO.
- Colour coded Kits : Based on
  - History + P/S examination + P/V examination + Minimal investigation.

Active space

Complaints	Presentation	Rx
Vaginal discharge	H/o : Discharge / pruritis / dysuria P/S : Discharge from vagina/ strawberry cervix P/V : Normal Cx : Healthy → Vaginitis	Green Kit 2 : • Secnidazole : 2 gms + • Fluconazole 150 mg stat
Discharge	H/o discharge P/S : Unhealthy • Erosion/ulcer/friable Cx → Cervicitis P/V : Normal	Grey Kit 1 : • T. Azithromycin : 1 gm OD stat + • T. Cefixime 400mg OD stat 
Lower abdominal pain	• H/o : Discharge/menstrual abnormality • P/V : Any 1 of the following : 1. Uterine tenderness 2. Cervical movement tenderness 3. Adnexal tenderness • R/o : Ectopic pregnancy	Yellow Kit 6

Supportive treatment :

- Abstinence x 7 days.
- Condom contraceptive.
- Partner Rx : Cervicitis & PID
  - Not indicated in vaginitis.

Other uses of Kit 1 :

- For partner Rx of cervicitis, PID.
- Urethritis.
- Anorectal discharge.
- Scrotal pain syndrome.

----- Active space -----

PELVIC INFLAMMATORY DISEASE (PID)

Features and Management of PID

00:01:33

Definition : Infection of upper female genital tract.

Includes :

- Endometritis (Uterus).
- Salpingitis (Fallopian tubes).
- Oophoritis (Ovaries).

Infection can spread to abdomen

1. Peritonitis.
2. Perihepatitis.

m/c route of spread : Ascending infection along with sperms (STDs).

m/c seen in : Sexually active females (15-35 yrs).

Risk factors :

- multiple risk factors
- H/o instrumentation → Post abortion. / Post partum.
- Previous H/o PID
- CuT insertion (1st 3 weeks).

PID : Polymicrobial infection.

Protective in PID

- Barrier contraceptives (Condoms).
- MIRENA.

Pathogenesis :

1. Uterus : Endometritis.
   - Menstrual abnormalities
   - Dysmenorrhea
   - Lower abdominal pain

   } Endometrial biopsy
2. Tubes :
   - Thick, congested, hyperemic, swollen tubes.
   - Endosalpingitis.
   - Pyosalpinx.
   - Hydrosalpinx.
3. Oophoritis.
4. Peritonitis & pelvic abscess.

m/c PID in :

- Reproductive age : Chlamydia > Gonorrhea.
- Virgins : Genital TB (Hematogenous)
- Menopausal females : E. coli.
- Cu - IUD users : Actinomyces.

mindmap no : 1

----- Active space -----

Symptoms :
- m/c :
  - Lower abdominal pain (B/L).
  - ↑Pain during coitus.
- Acute Phase :
  - Fever. → menorrhagia.
  - AUB → Intermenstrual bleed.
    → Post coital bleed.
- Dysmenorrhea.
- Infertility.
- Vaginal discharge.

Complication : Fitz-Hugh-Curtis Syndrome.
- Perihepatitis.
- Violin string adhesions : Between liver capsule & anterior abdominal wall.
- C/o right upper quadrant pain.
- m/c Chlamydia > Gonorrhea.

PID : Clinical diagnosis

CDC criteria for diagnosis :

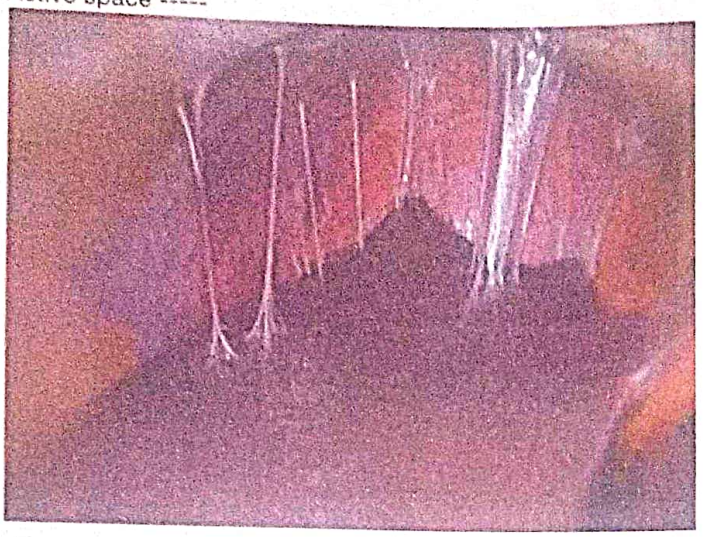
1. Minimal Criteria :
   Lower abdominal pain with any 1 of the following P/V findings :
   1. Cervical movement tenderness.
   2. Adnexal tenderness.
   3. Uterine tenderness.
2. Additional criteria :
   - Temp > 101°F.
   - Abnormal cervical /vaginal discharge.
   - Saline microscopy of vaginal discharge : Abundant WBCs.
3. Lab diagnosis of Chlamydial/Gonorrhea :
   - ↑ ESR
   - ↑ CRP

Investigations for definitive diagnosis :

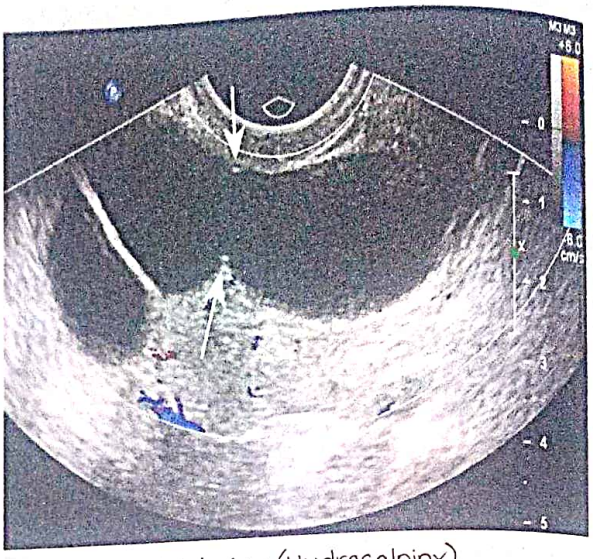
1. USG :
   - Tubo-ovarian mass .
   - Swollen tubes.
   - Signs :
     - Waist sign.
     - Bead on sting sign.
     - Cogwheel sign.
2. Endometrial biopsy : Endometritis.
3. Laparoscopy : Gold standard
   - Visualize tubes, adnexa & uterus directly.
   - Specimen collection.
   - Scoring for conception.
   - Rule out :
     - Ectopic pregnancy.
     - Appendicitis.

mindmap no. 2

----- Active space -----

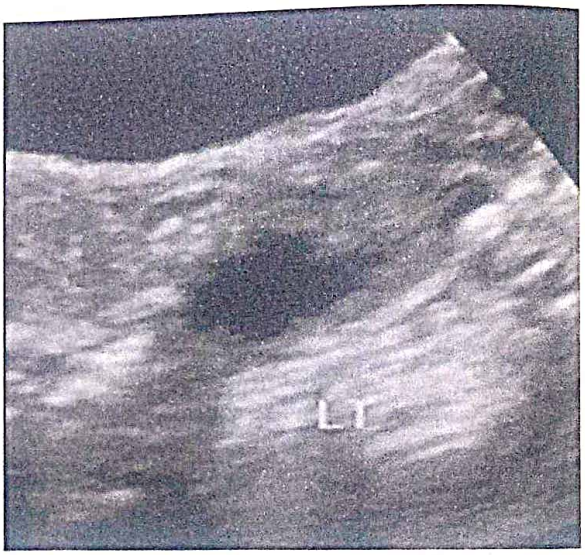


Violin string adhesions



Waist sign (Hydrosalpinx)

Beads on string sign



Cogwheel sign (Thickened endosalpingeal walls)

Note :

- Surest sign of PID : Pus trickling from fimbrial end during laparoscopy.
- Scoring system for conception in PID : Boer-Misel score (Type of organism not included).
- Staging of PID : Gainsville staging.
- Culpotomy : Drainage of pelvic abscess.

Beads on string → USG : PID
Beads on string → HSG : Genital TB

----- Active space -----

Management : OPD basis.
Broad spectrum antibiotics :
Inj. Ceftriaxone 250mg I.M single dose.
+
Tab Doxycycline 100mg BD x 14 days.
+
Tab Metronidazole 400mg BD x 14 days.

Indications for admissions :
1. Not responding to oral treatment.
2. Pelvic abscess.
3. Pregnant female with clinically severe disease.
4. Pregnant female unable to take oral medications.

Note :
I/V antibiotics continued for 24hrs after patient becomes afebrile .
+
Oral antibiotics x 14 days.

Long term consequences :
1. Infertility : m/c.
2. Ectopic pregnancy.
3. Chronic pelvic pain.
4. Hydrosalpinx.
5. Recurrent PID.
6. Fitz-Hugh Curtis syndrome.

Advice:
- Abstinence x 7-14 days.
- Partner treatment.
- Barrier contraceptive.
- Avoid alcohol (metronidazole).

PID

Syndromic approach

Advice :
- Abstinence x 7-14 days.
- Use of condoms.

Indications :
- C/o lower abdominal pain + any 1 of the following
  1. Uterine tenderness.
  2. Cervical motion tenderness.
  3. Adnexal tenderness
- Ensure UPT (-).

Treatment :
Patient :
No. 6 : Yellow Kit → Tab Cefixime 400mg stat.
→ Tab Metronidazole 400mg BD x 14days
→ Tab Doxycycline 100mg BD x 14days

Partner :
No. 1 : Grey Kit → Tab Cefixime 400mg single dose.
→ Tab Azithromycin 1g.

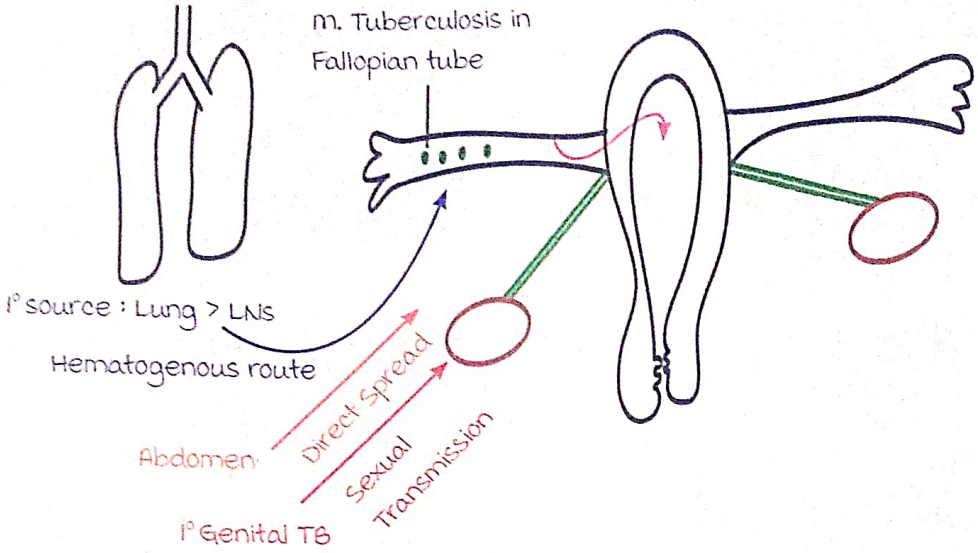
mindmap no. 3

----- Active space -----

GENITAL TUBERCULOSIS

PATHOGENESIS

- 2° infection.



Causative organism (mycobacterium tuberculosis)
↓ Enters
Fallopian Tube (FT)
↓ Lining destroyed from within
Endosalpingitis
↓
Caseous necrosis
↓
Fibrosis / Adhesion
↓ Enters uterus via cornual block
B/L cornual block

- From uterus :
  - Spreads to Endometrium : Fimbriae spread.
  - Causes adhesions.
  - Causes hydrosalpinx.

Salient Features :

- Spread → m/c route : Hematogenous.
  → Rare : Sexual Transmission.
- m/c sites : Fallopian tube (Ampulla) : 90-100% > Uterus : 50-60% > Ovary.
- L/c site : Vagina & Vulva (≤2%).

----- Active space -----

Genital TB of Various Sites

00:05:30

SITES

1. Genital TB of Fallopian Tube :

- B/L
- m/c site : Ampulla > Isthmus.
- m/c presentation : Infertility.

Investigations :

- IOC for tubal patency : HSG (Hysterosalpingography).
- HSG in Genital TB :
  - C/I : Can lead to dissemination.
  - Done after ATT (Anti-Tuberculosis Treatment).
- Characteristic Apperance of tubes on HSG :
  a. Beaded apperance.
  b. Lead pipe apperance.
  c. Tobacco pouch apperance.
  d. Golfstick apperance.
  e. B/L cornual block.
  f. Cotton wool apperance.
  g. Hydrosalpinx.

Beaded Apperance

Lead pipe Apperance

Tobacco pouch Apperance

Lead pipe Apperance

----- Active space -----

2. Genital TB of Uterus :

- m/c route of spread : Direct spread from fallopian tube.

Sites :

- Affected : Superficial layers of Endometrium.
- Spared : myometrium.
- Significance :
  - Ist day menstrual blood can be used for Investigation.
  - Best time for endometrial biopsy : Pre-menstrual phase (Tubercles in superficial layers)

Presentation :

- Initially : Endometrium normal → No menstrual symptoms.
- Acute : Endometritis → menorrhagia.
- Chronic : Adhesions (Asherman syndrome) → 2° amenorrhea/ Oligomenorrhea.

3. Genital TB of Ovaries :

- TB limited to : Surface of ovary by Tunica albuginea → Prevents spread to stroma.
- Appearance : Tubercles on surface of ovary (On laparoscopy).

4. Tubercular Peritonitis :

Genital TB associated with peritonitis → Ascites.

Feautres and Management of Genital TB

00:14:40

CLINICAL FEATURES

- m/c age group : 20-30 years.
- ± Seen in young girls.
- m/c cause of PID in virgin females.

Prodromal Symptoms :

- Fever (Low grade).
- Malaise.
- ↓ Appetite, weight loss.

Presentation :

- m/c : Infertility d/t Tubal blockage.
- menstrual irregularities
  - Initially : menorrhagia d/t Endometritis
  - m/c : 2° amenorrhea/ hypomenorrhea d/t Asherman syndrome.
- ± Chronic pelvic pain
- Rare : Serosanguinous (Blood stained) discharge.
- Future risk of : Ectopic pregnancy.

EXAMINATION

	Findings
General examination	Fever +/- weight loss.
P/A examination	Doughly Abdomen, Ascites.
P/S examination	Ulcers on cervix/vagina/vulva.
P/V examination	• Pelvis → Normal (m/c) / Frozen. • Adenexal mass → Reproductive age: Rare / Young / Pubertal females: B/L mass (m/c finding)

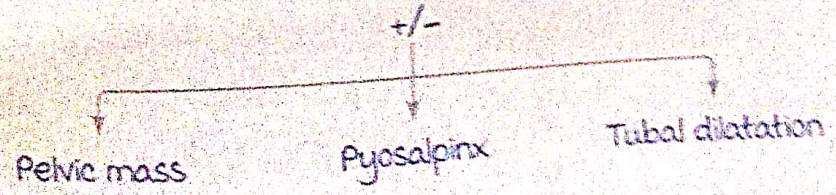
Findings are +/- → Dx of high suspicion.

DIAGNOSIS

History: → High suspicion
- Past H/O TB.
- Family H/O TB.
- Chronic PID: Unresponsive to broad spectrum antibiotics.
- PID in young/pubertal females.

Endometrial Biopsy:
- Sample site: Cornua (B/L).
- Phase: Premenstrual phase.
- Sample sent for:
  - Culture: In saline.
  - NAAT (Nucleic acid amplification Test).
  - PCR: Confirmative.
  - Histological Diagnosis: In formation.
- Virgin females: Endometrial biopsy should not be done. → Take 1st day menstrual blood.

USG:



----- Active space -----

Laparoscopy :

Site :

1. Fallopian Tube : Take sample → To confirm.
   - Red-hyperemic.
   - Swollen.
   - Dilated.
2. Ovary
3. Peritoneum

} ± Tubercles

Indication : High level of suspicion with -ve Endometrial biopsy.

MANAGEMENT

	Rx
TB	ATT (Category I) x 6 months : Can improve fertility.
Asherman Syndrome	Hysteroscopic adhesiolysis + Pediatric Foley's Catheter + (Estrogen + Progesterone) x 3 months
Infertilty	IVF + pelvic anatomy distorted/Tubal block.

FEMALE INFERTILITY : PART 1

----- Active space -----

CAUSES

- Ovarian cause (20-40%) : m/c cause.
- Tubal cause: 2nd m/c.
- Uterine cause :
  - Submucous fibroid
  - Endometrial polyp.
  - Asherman Syndrome.
  - Mullerian malformation : Septate uterus.
- Cervical & unexplained causes :
  - Anti sperm antibodies.
  - Cervicitis.

CLASSIFICATION (WHO) OF OVARIAN CAUSES

Class	FSH	Estrogen	Feature & Example
1	↓	↓	Hypogonadotropic hypogonadism Eg : Kallmann syndrome
2	Normal	Normal	Normogonadotropic normogonadism Eg : PCOS Anovulation
3	↑	↓	Hypergonadotropic hypogonadism Eg : Primary ovarian insufficiency (POI)/Premature menopause

- Hyperprolactinemia : Not classified (Separate cause) :
  - Applied aspect : Couple with c/o infertility → Males & females → S. Prolactin levels, S. TSH levels

Class I - Kallmann syndrome :

Mx : Pulsatile GnRH.

Class II - Anovulation

00:06:45

- m/c cause of ovarian infertility.
- Most easily treatable form of infertility.

Anovulatory cycles :

- Irregular cycles.
- D/t Estrogen breakthrough bleeding.
- Painless.
- Heavy menstrual bleeding.

Active space

TESTS FOR OVULATION

a. Based on history :

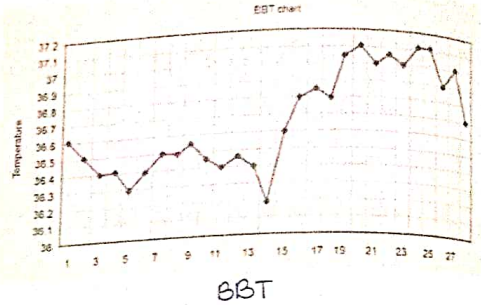
Retrospective test :

- Regular cycles.
- Dysmenorrhea.
- mittelschmerz syndrome.
- Premenstrual syndrome.

b. Based on progesterone :

- Done on day 21 of cycle : Highest concentration of progesterone (or) 1 week before menstruation.

1. Basal body temperature – BBT :
   - Progesterone : Thermogenic.
   - ↑ BBT by 0.5° to 0.8°F.
   - Biphasic temperature chart: Ovulation occurred.

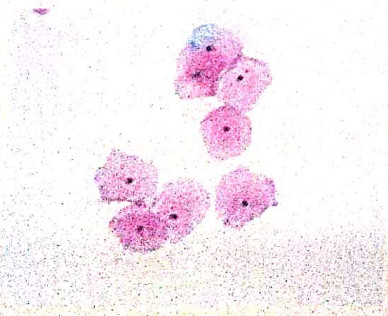


BBT

2. Cervical mucus study :

Anovulatory cycle	Ovulatory cycle
Estrogen (Predominant).	Progesterone (Predominant).
• Profuse, watery, elastic mucus.	• Thick, scanty, viscous mucus.
• Spinnbarkeit phenomenon: Stretchability.	• Cannot be stretched.
• Fern like appearance under microscope.	• Ferning absent.

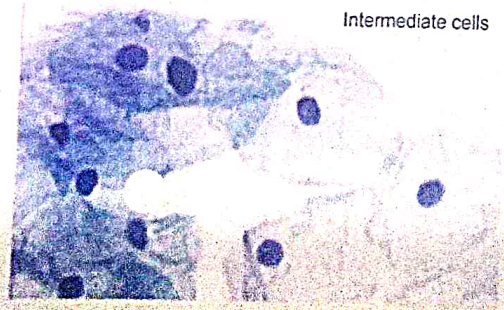
3. Vaginal epithelial cell study :
   - Specimen : Taken from lateral wall of vagina.



- Estrogen predominance.
- Eosinophilic superficial cells.

↓

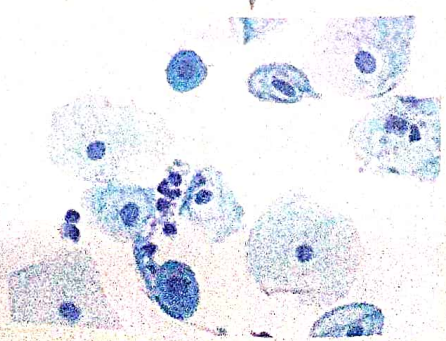
Ovulation –



- Progesterone predominance.
- Basophilic – Intermediate cells.

↓

Ovulation +



- No hormonal Predominance.
- Parabasal cells : Blurred margins with big nucleus.

----- Active space -----

Maturation index :

- Counting of 100 cells.
- A/B/C = No. of parabasal cells/No. of intermediate cells/No. of superficial cells

Ovulation :

Eg :

- 0/10/90
- B predominates

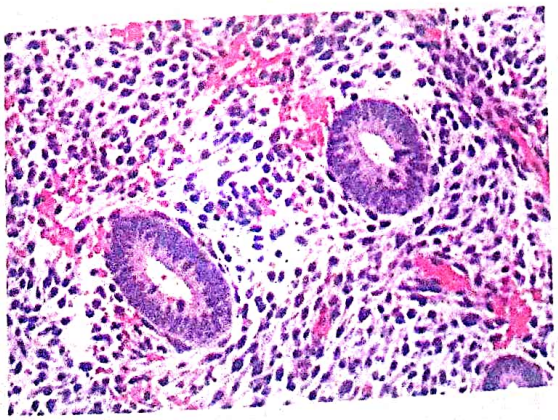
Non-ovulation :

- 0/40/60
- C predominates

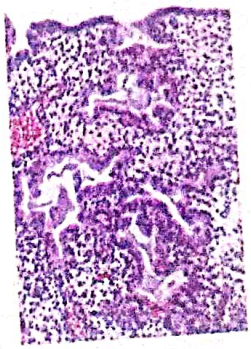
Karyopyknotic index :

- $\frac{\text{Superficial cells}}{\text{Intermediate \& parabasal cells}}$
- max : During ovulation.
- Depends on estrogen levels.

4. Endometrial biopsy :
   - Done on day 26 of cycle, in premenstrual phase



Single tubular glands
D/t estrogen
Proliferative phase
↓
Ovulation –



- Corkscrew/Saw-tooth glands :
  - Filled with secretion.
- D/t progesterone.
- Secretory phase.

↓
Ovulation +

- Luteal phase defect :
  - Difference of ≥2 days between history & biopsy.
  - ↓ Progesterone.
  - ↑ Chance of abortion during pregnancy.
- Not done solely to test ovulation.

----- Active space -----

- Indication :

a. Genital TB diagnosis :
Sample collected → Histopathology : Formalin
→ Acid-fast bacilli : NS

b. AUB.

5. Serum Progesterone levels :
- Simple, easy to perform, cheapest test.
- Interpretation :
  - ≥3 ng/mL → Ovulation +
  - Max: 15 ng/mL (6-8 days after ovulation).

TESTS WHICH PREDICT OVULATION

1. Urinary LH Surge :
- Occurs 24 hrs before ovulation.
- Tested with kits.

Note :
Serum LH surge: 36 hrs before ovulation.

Drawbacks :
- Does not specify about ovulation : LH Surge does not guarantee ovulation.
- LH levels constantly raised in PCOS → Test cannot be performed.

2. Follicular monitoring :
- m/C done test.
- Done from day 10, on alternate days.
- Studied by transvaginal sonography (TVS).
- Size of follicles ↑ 2 mm/day → Up to 18-20mm → Size ↓ & becomes irregular.

Signs of ovulation :
- ↓ in Size of follicle.
- Irregular follicle.
- Fluid in Pouch of Douglas d/t rupture of antral follicle.
- USG : Triple layered/Trilaminar endometrium : Seen in late proliferative > Ovulatory phase.

Triple layered endometrium

MANAGEMENT

Ovulation inducing drugs.

DOC for :
- Anovulation : Clomiphene citrate.
- Anovulation :
  - D/t PCOS : Letrozole > Clomiphene citrate.

- D/t pituitary ablation : HMG.
- D/t Kallman syndrome : Pulsatile GnRH.
- DOC for hyperovulation in patients undergoing IVF : HMG (GnRH).

Class III - Premature Menopause/POI

No ovarian follicles in females <40yrs.

Follicle

Granulosa cells :
- Inhibin B } -ve feedback on FSH
- Estrogen }
- AMH (From menarche to puberty)

Theca cells :
Androgen

TESTS FOR OVARIAN RESERVE

Tests to check if sufficient follicles are present in ovary.

Tests done on day 2 – day 3 of cycle :

1. Inhibin :

↓ Ovarian reserve ⟶ ↓ Inhibin B < 45 pg.

2. FSH :
- m/c done test.
- ↓ Ovarian reserve ⟶ ↓ Inhibin, ↓ Estrogen ⟶ ↑ FSH (Loss of -ve feedback)

Inference :

FSH	Inference
2–10 IU	Normal
10–15 IU	↓ Ovarian reserve
>25 IU	Suggestive of POI
≥40 IU	Diagnostic of POI

Drawback : Done on day 2 – day 3 of cycle.

3. Antral follicle count (AFC) :
- No. of antral follicles counted in both ovaries by TVS.
- Antral follicles : 2 – 9 mm ; AFC : < 10 ⟶ POI

Tests done on any day of cycle :

Anti Mullerian Hormone (AMH) :
- Glycoprotein hormone.
- Released by pre-antral follicles (mainly) + Small antral follicles.
- Best test.

----- Active space -----

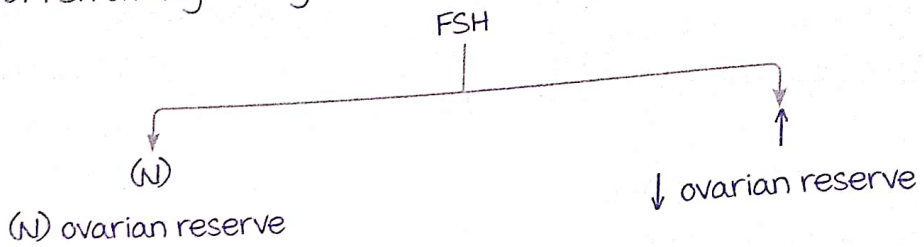
Inference :

AMH level (ng)	Inference
1-3	Normal
< 1	Borderline reserve
< 0.5	Poor reserve : POI
> 3	High : PCOS

Other tests :

Clomiphene citrate challenge test : Obsolete.

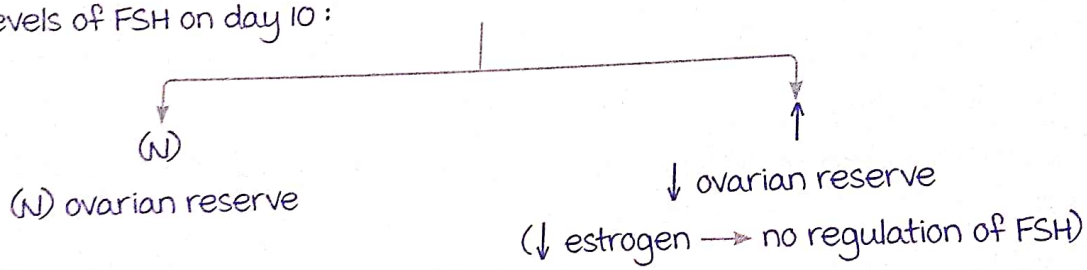
a. Levels of FSH on day 3 of cycle :



b. Clomiphene citrate (CC) between day 5 - day 9 :

CC → ↓ Estrogen → ↑ FSH → multiple follicles mature → Estrogen released → FSH regulated.

c. Levels of FSH on day 10 :



Tests not done for ovarian reserve :

1. S. Progesterone: test for ovulation.
2. S. Estradiol.

Note : Investigations for infertility in couples :

- Semen analysis.
- Test for ovulation : S. Progesterone.
- HSG.

Indications :

- Females age ≥35 yrs.
- Family H/o POI.
- H/o Chemotherapy/Radiotherapy.
- Unexplained infertility.

MANAGEMENT

- C/o Amenorrhoea/Hot flushes : HRT.
- C/o Infertility : IVF with donor eggs.

FEMALE INFERTILITY : PART 2

----- Active space -----

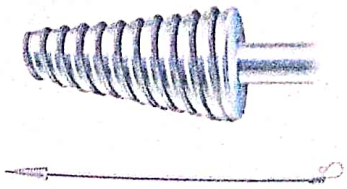
Tubal Causes

00:00:17

- PID
- Salpingitis
- Endometriosis

→ Tubal blockage → U/L, B/L

TESTS FOR TUBAL PATENCY



Leech Wilkinson Cannula

HSG :

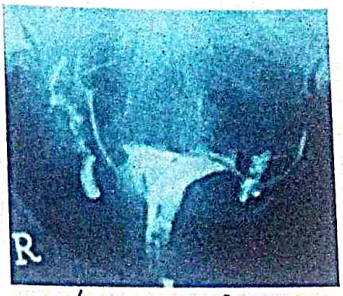
- IOC
- Radiopaque dye injected into uterus.

↓

Serial X-rays

↓

Assessment for spillage.



B/L spilling of dye

Time of HSG : Day 7- 10 of cycle.

Drawback :

- Exterior of tube is not visualised.
- Painful Procedure → Spasm of cornual ends of tubes → False +ve tubal blockage.

Note :

- Basic investigation in all couples :
  - Female : HSG + ovulation tests.
  - Male : Sperm analysis.
- In females > 35 yrs : Tests for ovarian reserve.

Laparoscopic chromopertubation :

- Gold standard.
- Dye : methylene blue/Indigo Carmine.

Procedure :

- Laparoscope is inserted.
- Dye is passed through the cannula.
- Spillage of dye is directly visualised.

Advantage :

Exterior of tubes can be visualised.

Dye in Laparoscopic chromopertubation

----- Active space ----- **TUBAL BLOCK**

- U/L
  - Mx like unexplained infertility : Clomiphene citrate (+) IUI
- B/L on HSG
  - Confirmation by lap. chromopertubation
    - B/L proximal/cornual block
    - B/L distal/fimbrial block

B/L Proximal/Cornual Block :

a. Physiological Spasm : m/c cause.

↓ Next step

Hysteroscopic cannulation :

Guide wire passed via uterus into fallopian tube(FT) under hysteroscopic guidance.

1. Release of cornual spasm
2. Dislodge mucus plug.

↓ Next step

Lap. chromopertubation

b. Genital TB :

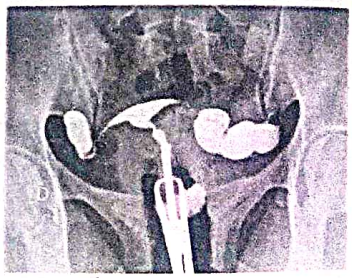
- m/c pathological cause.
- Mx : IVF.

B/L cornual block

B/L Fimbrial block :

On Lap. chromopertubation.

1. Confirmation of Dx.
2. Assessing extent of disease :
   - Mild disease → Fimbrioplasty : Small nicks in fimbrial ends → Opens fimbria.
   - Severe disease : Hydrosalpinx → IVF



B/L Fimbrial block : Hydrosalpinx

Note : management of choice in tubal infertility → IVF.

Hydrosalpinx :

- Collection of secretions in fallopian tube.
- Fluid : Fetotoxic.
- management → (Clip applied at proximal end / or / Salpingectomy.) → Followed by IVF.

Mid segmental block :
- Block in middle part/isthmus.
- Site of tubal ligation (Permanent/irreversible method).
  - Needing recanalization (Done in exceptional circumstances).

Success of reversal depends on :
- Age of female < 35yrs.
- Type of anastomosis : Isthmo - isthmic > isthmo - ampullary anastomosis.
- Length of tube after reconstruction : > 4 cms.
- Type of procedure done for tubal ligation : Hulka clips > Fallope rings.

Note : Chances of reversal does not depend on interval between tubal ligation & recanalization (Unlike vasectomy).

Uterine & cervical causes

00:31:20

UTERINE CAUSES

Causes :
- Fibroid
  - m/c : Submucous fibroid.
  - mx : myomectomy.
- Endometrial polyps
- Asherman syndrome.
- Chronic endometritis d/t chlamydia.
- Diethylstilbestrol (DES) exposure in utero → T shaped/Hypoplastic uterus.

Note :
- Mullerian malformation leads to Recurrent pregnancy loss (m/c), not infertility.
- Septate uterus : may lead to infertility.

Investigation :
- USG
- Hysteroscopy : Gold standard. } Coincidental finding : Mullerian malformation.

CERVICAL CAUSES

- Cervicitis
- Cervical stenosis
- Antisperm antibodies.

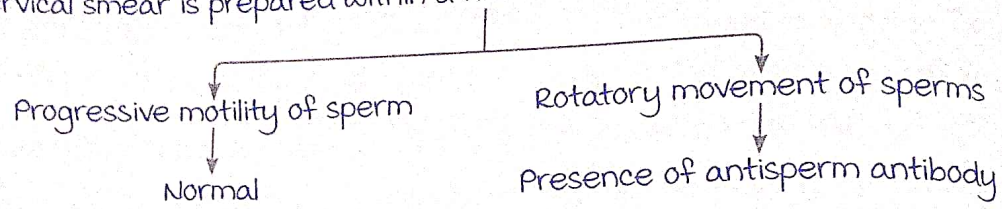
Antisperm antibodies in females :
- Spermicidal → Infertility.
- AKA Immunological cause of infertility.

----- Active space -----

Investigation :

Post coital/Huhner test : Obsolete

- Detects antisperm antibody.
- Done on Day 12 – Day 14.
- Cervical smear is prepared within 2 hrs of intercourse.



Management :

Clomiphene citrate + IUI (Intrauterine insemination)

- Also the Rx of unexplained infertility.
- Therefore, test for ASA not done.
- Cervical causes included in cause of unexplained infertility.

Clomiphene citrate :

- Started any day from D2 – D5.
- Duration : 5 days (m/c : D5 – D9).
- On day 10 : Follicular monitoring → Follicle ≥ 17 mm

↓

Inj. HCG : Ovulation trigger (Not absolutely necessary)

↓ 32 – 36 hrs

IUI

Miscellaneous

00:42:00

HyCoSy :

- Hystero-contrast sonography.
- Injection of contrast enhancing agent into uterus :

Lignosol : 2% Lignocaine jelly + saline

↓

Passage through tubes visualised by USG

Advantages :

- OPD procedure
- Less painful
- Safe
- Cost effective.

Rubin's cannula :

- Used for Rubin's test/CO2 insufflation test.
- Obsolete test for tubal patency.

MALE INFERTILITY

Active space

Infertility

00:00:32

Definition :

Inability to conceive even after 1 yr of unprotected intercourse.

Age of investigating infertility :

Age of female	Period of unprotected intercourse
< 35yrs	1yr
≥ 35yrs	6 months
> 45yrs	3 months

Types :

1° infertility : Female has never conceived.

2° infertility :

- Female had conceived (irrespective of outcome).
- Unable to conceive currently.

Causes :

Female factors : 40–35%

male factor contribution : 40%.

male factors (Alone) : 20%.

unexplained : 10%.

Note :

Fecundability	Fecundity
Probability of conceiving in one cycle.	Probability of having a live birth in one cycle.
20% in the first cycle after marriage.	50% in the first cycle after marriage.

Investigations :

Basic investigations :

1. Semen analysis.
2. Test for ovulation : S. Progesterone levels on day 22 of cycle
3. Test for tubal patency : HSG between day 7 – day 10.

} If normal → Unexplained infertility

↓

mx : Clomiphene citrate + IUI.

Female partner ≥35yrs : Test for ovarian reserve → Serum AMH.

Causes & Evaluation of Male Infertility

00:09:28

Common causes :

- m/c cause : Abnormal spermatogenesis d/t testicular causes → Gonadal failure.
  Eg : Klinefelter syndrome.
- m/c reversible cause : Varicocele.

Note : ↑ Paternal age → Structural & chromosomal defects in sperms.

Quality of sperms monitoring – Sperm DNA fragmentation test : Done before IVF.

----- Active space -----

SEMEN ANALYSIS

- After 3 days of abstinence (2-7 days).
  - < 2 days : False low sperm concentration.
  - > 7 days : False high dead sperm concentration.
- Sample should reach lab within 1 hr.
- Analysis is done on liquefied semen → Liquefaction time : 20 to 30 mins.
- Sample collection :
  - By masturbation into a sterile container.
  - Not collected in latex condom.
- Maturity & motility of sperms :
  - Maturity : Proximal end of epididymis.
  - Motility : Distal end of epididymis.

WHO 2022 Semen Analysis Parameters :

Features	Minimum cut off	Conditions
Volume	1.4 mL	Aspermia : Absence of semen
pH	7.2	-
Concentration	16 million/ml	• Oligospermia : < 16 million/ml • Severe oligospermia : < 5 million/ml • Azoospermia : No sperms → Hormonal profile + Genetic testing → To rule out Klinefelter syndrome
morphology	4% as per Strict Tygerberg criteria	Teratozoospermia : ↑ in abnormal sperms
motility	• Total motility : 42% • Progressive motility: 30%	Asthenospermia: ↓ Sperm motility
Viability	54%	Necrozoospermia: ↑ in no. of non viable sperms
Total sperm count	39 million/ejaculate	-

Most important parameters : morphology > motility > Concentration.

Sperm parameters constant from 2010 to 2022 :

- Sperm morphology.
- Total sperm count.

Common causes of azoospermia :

- Klinefelter syndrome : 47XXY.
- Cystic fibrosis :
  - Congenital B/L absence of vas deferens & seminal vesicles.
  - CFTR gene mutation.

Common causes of asthenospermia :
- Kartagener syndrome/Immotile cilia syndrome : c/o infertility + Recurrent sinusitis + Recurrent chest infection d/t Bronchiectasis.
- Smoking.

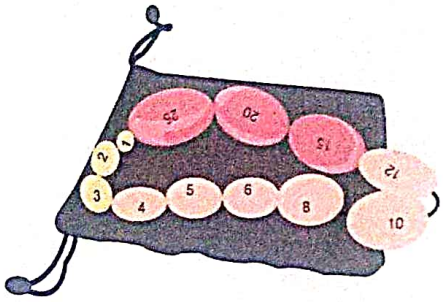
Note : Young's syndrome → Mucus thickening → Blockade of nose, bronchi & vas deferens. → c/o infertility, sinusitis, bronchiectasis.

OATS :
- Oligospermia + Asthenospermia + Teratozoospermia.
- Seen in varicocele & smoking.

KLINEFELTER SYNDROME

- 47XXY
- Incidence : 1 – 2.5/1000 pregnancies.

Characteristics :
- Tall, long limbs, gynaecomastia
- Testis : Small, hypoplastic.
- ↑Risk of Breast cancer, Non-Hodgkin's lymphoma, Extragonadal germ cell tumour
- ↓Life span, ↓IQ.



Prader orchidometer
(measures testis size)

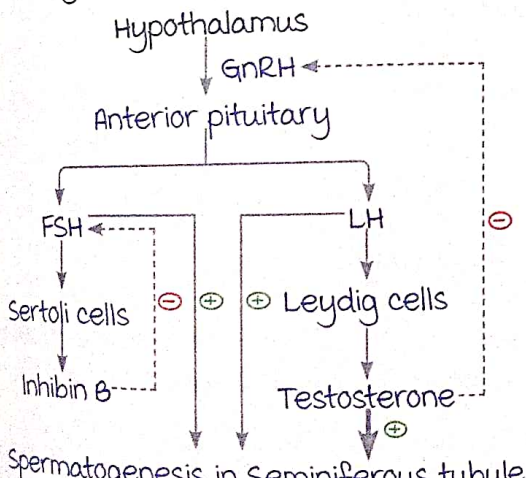
Spermatogenesis & Sperm pathway

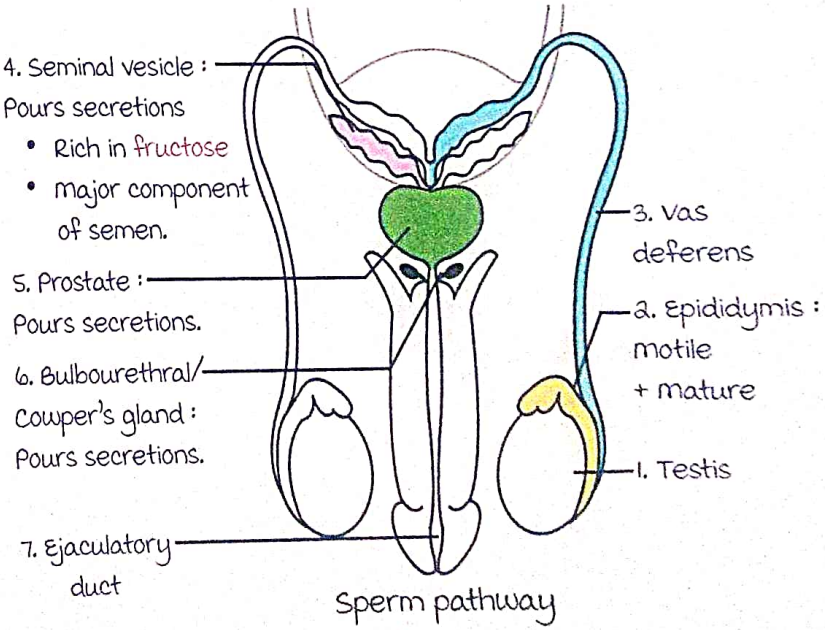
00:28:12

- Defect in spermatogenesis.
- Obstruction in sperm passage.

→ Infertility in males.

Spermatogenesis : Begins at puberty.





Sperm pathway

----- Active space -----

Levels of organs involved in spermatogenesis :

Pre-testicular organs : Hypothalamus & pituitary.

Testicular Organs : Testis

Post-testicular organs :

- Seminal vesicles, Epididymis, Ejaculatory duct, Vas deferens.
- Not necessary for spermatogenesis.
- Necessary for sperm passage.

Semen :

- Seminal vesicle.
- Prostate gland : Prostatic enzymes → Liquefaction of semen.
- Bulbourethral gland : Adds secretion at the time of intercourse.

Diagnosis of Azoospermia

00:40:06

AZOOSPERMIA ON SEMEN ANALYSIS

Azoospermia in Ist semen analysis.

↓

Repeat semen analysis. → • Ideal : After 1 month. • min : After 7 days.

↓

Azoospermia.

↓

FSH, LH, Testosterone levels.

↓FSH & ↓Testosterone	↑FSH & ↓Testosterone	Normal FSH & Normal Testosterone
↓	↓	↓
Hypogonadotropic hypogonadism.	Hypergonadotropic hypogonadism.	Obstructive azoospermia.
↓	↓	↓
Hypothalamus/ pituitary causes	Testicular cause : m/c	Obstruction in Sperm pathway
↓	↓	↓
1. Kallman syndrome. 2. Hyperprolactinemia d/t -ve feedback on GnRH. 3. Chronic hypothyroidism : ↓T_3 → ↑TRH → ↑TSH, ↑Prolactin.	1. Klinefelter syndrome. 2. Testicular tumor 3. Mumps orchitis. 4. ↑Heat exposure (miners).	1. Varicocele. 2. Cystic fibrosis. 3. Vasectomy 4. Retrograde ejaculation. 5. Genital TB

Obstructive azoospermia :

Low semen volume,
↓Fructose
No contribution from seminal vesicle.
Trans rectal ultrasound
Seminal vesicles

Not seen.
Cystic fibrosis : Absence of vas deferens & seminal vesicle.
Confirmation : CFTR gene mutation.

Seen.
Dilated.
Ejaculatory duct obstruction.

Normal.
Retrograde ejaculation.

Normal semen volume,
Fructose +
Contribution of seminal vesicle to semen.
Block above seminal vesicle
- Vas deferens.
- Epididymus.

Scrotal USG

----- Active space -----

Treatment Procedures

01:00:57

INTRAUTERINE INSEMINATION

Injecting sperm into uterus.

Prerequisite :

- Sperm concentration : 10 million/ml.
- Sperm morphology : 14%.

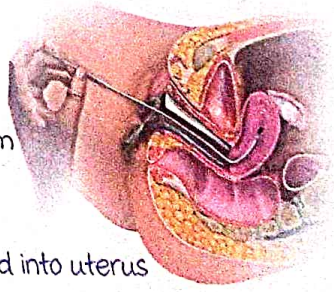
Obtaining semen :

- From husband : IUI-H.
- From donor : IUI-D.

Processing of semen :

- Swim up technique.
- Swim down technique.
- Density centrifugation technique.
- Glasswool filtration technique : Best.

Sperm in sterile medium



Sperm injected into uterus

Intrauterine Insemination (IUI)

Procedure :

0.5 mL of conc. processed sperm.
↓
Injected into uterus with IUI catheter, bypassing cervix.
↓
Sperm travel to fallopian tube.

----- Active space -----

Done in :

- Cervix abnormalities.
- Ejaculatory abnormalities.

Not done : In females with tubal blockage.

Note : Assisted Reproductive Techniques

- Includes :
  - IVF
  - ICSI
  - GIFT
  - ZIFT
- Does not include IUI.

Indications :

- Oligospermia : 10-16 million/mL.
- Ejaculatory causes : Hypospadias.
- Retrograde ejaculation.
- Vaginismus : Involuntary contraction of vaginal muscles.
- Discordant HIV status in the partners.
- Anti sperm antibodies in cervix.
- mild endometriosis.
- Unexplained infertility.

Note :

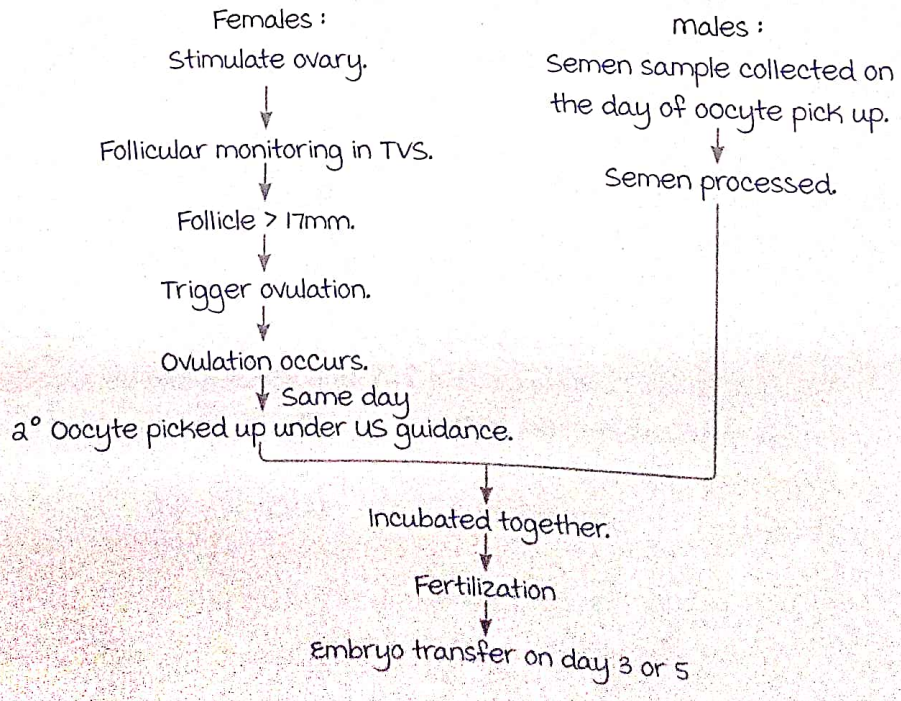
- IUI done at the time of ovulation.
- HCG is done before IUI to rule out tubal blockage.

IN VITRO FERTILISATION

Pre-requisite :

Sperm count : 5 million/mL.

Procedure :



Female ovarian stimulation :

• Controlled ovarian hyperstimulation :
  - Success rate of IVF : 20–30%.
  - Hyperstimulation → multiple 3° oocytes → 2–3 embryo implantation.
• Drug used : HMG/GnRH analogue.

Monitoring :

• Follicles (on TVS) : Numbers, size.
• Serum Estradiol levels.

Ovulation trigger :

• Done when ≥ 2 follicles with ≥ 17mm.
• Drug used : Inj. hCG.
• Ovulation occurs 33–36 hrs after Inj. hCG.

Oocyte retrieval : Done under US guidance in OT, under anaesthesia.

Fertilisation : On the same day as retrieval.

Culture media. : Semen + oocyte added.

• Rich in pyruvate.
• 50,000 – 1,00,000 sperms per oocyte.
• Fertilisation occurs in incubator, in petri dish.

Embryo transfer :

• On → Day 3 : Cleavage stage.
  → Day 5 : Blastocyst stage (Better).
• Site : 2 cm below uterine fundus.
• Instrument : Cooks catheter.
• Progesterone given prior to embryo transfer → prepares endometrium for implantation.

Pre-implant genetic testing : Done on day 3/day 5 embryo.

↓ No abnormality

Implanted.

Cryopreservation : Can be done for secondary oocyte.

Success rate : 20–30% → ↑ed by multiple embryo transfer.

Indications :

• Oligospermia : 5–10 million/mL
• Tubal cause of infertility.
• IVF with donor eggs in premature ovarian insufficiency
• IVF with surrogacy in mullerian agenesis.
• Severe endometriosis
• Pre-implantation genetic testing.

Not used in :

• Severe oligospermia.
• Azoospermia.
• Asthenospermia.

----- Active space -----

INTRACYTOPLASMIC SPERM INJECTION

Procedure :

- Collection of 2° oocyte & semen same as IVF.
- Single sperm injected into cytoplasm of oocyte → 1 sperm/oocyte.

Indications :

- Severe oligospermia : < 5 million/ml.
- Azoospermia.
- Asthenopermia : motility < 5%.

Testicular azoospermia :

↑FSH, ↑LH, Testosterone↓ → Testicular biopsy :
To identify presence of sperms in testis.
↓ Sperm+
Surgically extracted
↓
ICSI

Surgical methods to retrieve sperms :

TESA : Testicular Sperm Aspiration.

TESE : Testicular Sperm Extraction.

PESA : Percutaneous Epididymal Sperm Aspiration.

MESA : Microsurgical Epididymal Sperm Aspiration.

m_x OF SPECIFIC CAUSES OF MALE INFERTILITY

Condition		Treatment
Oligospermia	10–15 million/mL	IUI
	5–10 million/mL	IVF
	< 5 million/mL	ICSI
Non obstructive azoospermia		Surgical extraction of sperms → ICSI
Erectile dysfunction		Sildenafil
Premature ejaculation		SSRI
Ejaculatory dysfunction (Hypospadias)		IUI
Retrograde ejaculation		Sperms taken from urine → IUI

Note :

- Difference between dead sperm & live sperm :
  1. Supravital stain (Eosin/Trypan blue) → Stained : Dead sperm. / Not stained : Live sperm.
  2. Hypotonic osmotic sperm swelling test → No swelling : Dead. / Swells : Live, immobile.
- Round cells on semen analysis :
  myeloperoxidase test (Endtz test) → +ve : WBC. / -ve : Round sperm without an acrosomal cap.

CONTRACEPTION : PART 1

Active recall

Oral Contraceptives

00:01:45

mechanism :

Contraceptive	Main mechanism	Other mechanism
1. OCPs (E + P)	Anovulation (↓ FSH >>> ↓ LH)	• Thicker cervical mucus • Thinning of endometrium • ↓ Tubal motility
2. POP (contains LNG)	Make cervical mucus thick	
3. Progesterone implant/Injection/cerazette (Contains Desogestrel)	Anovulation	Inhibits implantation
4. Cu IUCD	Prevents fertilization (Cu : induces inflammatory reaction)	Inhibits implantation
5. MIRENA	Prevents implantation	-
6. Centchroman (Chaya)	Inhibit implantation by making endometrium out of phase	-

CONTRACEPTIVES - GENERAL

Long-acting reversible contraceptives (LARC) :

1. Injection : DMPA
2. Implants
3. IUCD

Permanent methods :

1. Tubal ligation
2. Vasectomy

Epilepsy & contraception :

1. Not on any antiepileptic : Any contraceptive can be used (OCPs also)
2. Patient on antiepileptic
   - Progesterone only contraceptive used
   - OCPs C/I (↓ efficacy of OCP's)

Note : OCPs ↓ efficacy of lamotrigine (↑ chance of seizures).

DMPA Injection (Depot medroxy Progesterone Acetate) :

Included in NFP : Antara

Dose : 150 mg, repeated every 3 months

Route : Intramuscular

missed dose :

- <4 wks of due date : Can be given upto 2 wks before/4 wks after due date.
- >4 wks after due date :
  1. Rule out pregnancy (UPT).
  2. Injection given + advise backup contraceptive x 7 days.

Mechanism of action : Anovulation.

Advantages : ↓ seizure frequency in epileptics.
↓ sickling in sickle cell anemia.

Disadvantage :

- Delayed return of fertility (12-18 months)
- ↓ bone mineral density
- Irregular bleeding : m/c reason for discontinuation

----- Active space -----

Classification & Contraindications of OCPs

00:19:19

Classification based on estrogen dose :

High dose : (Ethinyl estradial) EE ≥ 50 mcg

Low dose : EE < 50 mcg (30 mcg)

Very low dose : EE < 20 mcg

Minimum possible dose : 10 mcg

↓

α oesthin pill

Classification based on progesterone :

1st generation : Norethindrone

2nd generation : LNG

3rd generation : Least androgenic side effects

- Desogestrel
- Norgestimate
- Gestodene

4th generation : Drospirenone

- Antiandrogenic
- Antimineralocorticoid → weight loss
- ↑ K+

OCPs

Absolute C/I of OCP's :

Mnemonic : Banks have various schemes to provide home loans during may.

- H/o Breast cancer.
- Severe uncontrolled hypertension → WHO : ≥160/100 mmHg; CDC : ≥ 140/90 mmHg
- H/o Venous/Arterial thrombosis/MI/CAD/stroke/lupus anticoagulant antibody (+).
- Smokers ≥ 35yrs of age.
- Pregnant / h/o peripartum cardiomyopathy/< 21 days postpartum.
- Hypertrigliceridemia.
- Abnormal liver function/Hepatic adenoma/Liver cancer / h/o cholestasis during pregnancy.
- Diabetes with vasculopathy.
- Migraine with aura.

Note :

1. m/c side effect of OCPs : Irregular/Breakthrough Bleeding (m/c with low dose pills d/t less 'E').
2. Elective surgery with anticipated immobilization : OCPs stopped 4 wks prior.

Regimen for OCP Consumption & Missed Pills

00:30:20 ----- Active space -----

Method of consumption :

- Can be started at any time.
- Ideal : D1 – D5 of cycle.
- >D5 : Backup method x 7 days.
- 1 pill everyday, at same time.

28 Tab pack :

- 21 Hormonal tablets.
  +7 Ferrous fumarate tablets.

- menstruation (+) : Fresh pack started.

21 Tab pack :

- 1 pill daily x 3 weeks & 1 week off
- 1 day of menstruation : Fresh pack started.

Regimens :

21/7 : 21 days pills + 7 days off.

24/4 : 24 days hormonal pills + 4 days off.

- Advantages → Better efficacy.
  → Lesser side effects.

Multiphasic pills : Pills with different doses in the same pack (obsolete).

OCPs

Return of menses : Post pill amenorrhea +/-

↓

D/t thinning of endometrium by progesterone.

Return of fertility : Rapid, ≤ 3 months.

Government of India guideline for missed pills :

missed 1 or 2 pills :

- One pill to be taken ASAP or 2 pills at scheduled time.
- Backup contraceptive method not required.

missed ≥ 2 pills :

- Backup method x 7 days.
- Unprotected intercourse in previous 72 hrs : Emergency contraceptive pill.
  - 1st/2nd week :
    - 1 pill to be taken immediately + complete pack as per schedule.
    - Backup method x 7 days.
    - H/o intercourse in previous 72 hrs : Emergency contraceptive pill.
  - 3rd week :
    - 1 pill to be taken immediately + complete pack as per schedule.

    +

    Immediately start new pack (non-hormonal pills omitted).

----- Active space -----

Non-Contraceptive Uses, Side Effects & Cancers Associated with OCs

00:47:20

Non-contraceptive uses :

1. Irregular cycles : DOC
2. Hirsutism : DOC
3. Acne with ↑ androgens
4. Atypical uterine bleeding (Fibroid/ Endometriosis/Polyp)
5. Cycles made anovulatory, hence used in :
   - Dysmenorrhea
   - Endometriosis
   - Mittelschmerz syndrome : Pain during ovulation
   - Menstrual migraine
   - Ovarian cysts
6. Estrogen + progesterone
   - HRT (Hormone Replacement Therapy)
   - Turner's syndrome

Advantages :

Easy return of fertility

Failure rates :

- Typical : 0-8%
- Perfect use : 0-1%

Adverse effects :

- Nausea, vomiting, bloating
- Breast tenderness
- Breast cancer : ↑ risk (d/t progesterone)
- Venous thromboembolism (D/t estrogen) : ↑ risk in smokers
- M.I. : ↑ risk in smokers
- Stroke : ↑ risk in smoker/hypertensive patients
- Candidiasis/Chlamydia infection : ↑ risk. (↓ overall risk of PID d/t progesterone.)

OCPs

Cancers :

↑ risk (reversible)
- Breast cancer
- Cancer cervix
- Hepatic adenoma

↓ risk
- Endometrial cancer (d/t progesterone)
- Ovarian cancer
- Colon cancer
- Benign breast disease
- Ovarian cyst

No effect
- Liver cancer
- Gall bladder cancer

Intrauterine Contraceptive Device (IUCD)

00:56:40 ----- Active space -----

Absolute C/I :

1. Undiagnosed vaginal bleeding (Ca cervix/endometrium).
2. Severe distorsion uterine cavity (Fibroid/septate uterus).
3. Active pelvic infection (Including cervicitis).
4. Known/suspected pregnancy.

Specific C/I :

- CuT : Wilson's disease
- Mirena :
  - H/o breast cancer.
  - Active liver disease.

Missing Thread :

1st step : TVS.

↓ not visible

Serial abdominal X-rays.

- **Coiled Thread :**
  - No intervention required.
  - or Hysteroscopy removal (If desired).
- **Perforation :**
  - IUCD in peritoneal cavity → Urgent laparoscopy/ Laparotomy for removal.
  - IUCD partly in peritoneal cavity → Hysteroscopy +. Laparoscopy.
- **Malpositioned :**
  - LUS.
  - Myometrium.
  - Cervix : Remove IUCD.

} nil intervention

Pregnancy + IUCD in situ :

- **Continuation of pregnancy not desired**
  - → Termination of pregnancy + IUCD removal
- **Continuation of pregnancy desired**
  - **Thread visible** → Remove IUCD + antibiotics
  - **Thread not visible** → IUCD left insitu + pregnancy continued

Risk to pregnancy :

- Infection
- Abortions
- Pre-term labor
- IUGR

No risk of teratogenicity

----- Active space -----

Usage of IUCDs :

Nulliparous women : Frameless IUCD

Patient with heavy menstrual bleeding : mirena

Patient on anticoagulant : mirena

H/o Thrombosis : CuIUCD/mirena

PID :

- 1st three weeks : ↑ risk of PID
- m/c PID : Actinomyces
  ↓ managed by
  IUCD removal + antibiotics
- Other PID :
  Oral antibiotics x 48 – 72 hrs → not responding → switch to iv + IUCD removal after 24 hrs

01:09:45

Implants

IMPLANTS

Types :

(1) Implanon :
- Composition : Etonogestrel
- Single rod : 40mm x 2mm
- Efficacy : 3 yrs

(2) Nexplanon :
- Similar to Implanon
- Radiopaque

(3) Norplant :
- 6 rods
- 0.5% failure rate

(4) Jadelle Norplant 2

(5) Capronor :
- Biodegradable
- Efficacy : 1 yr

(3) & (4):
- Obsolete
- LNG
- Efficacy : 1 yr

Implanon :

Releases : 60 mcg/day x 5 yrs

Approved for : 3 yrs

Insertion & removal : OPD procedure, medial side of non-dominant hand

Mechanism of action : Anovulation

m/c side effect : Irregular bleeding

Failure rate : 0.1% (most effective)

CONTRACEPTION : PART 2

----- Active space ----

Progesterone Only Pills (POP) & Tubal Ligation

00:00:25

Progesterone containing contraceptives :

mechanism of action :

- POPs/minipill : Renders cervical mucus thick.
- Desogestrel containing POP.
- Implants/DMPA injection.

→ Anovulation

Side effect : Irregular bleeding.

Safe in :

1. Breast feeding females.
2. Smoker.
3. H/o thrombosis.

Disadvantages of DMPA :

1. Delayed return of fertility.
2. Bone loss.

DOC in :

- Lactating women : POP > IUCD.
- ≥ 40yrs : Progesterone > OCP's.

POPs

Dose of POPs :

	minipill (LNG)	Desogestrel (Cerazette)
Dose	35 mcg	75 mcg
Window period	3 hrs	12 hrs

- Should be taken at the same time, daily.
- Delay in taking POPs : Backup method x 48 hrs.

----- Active space -----

TUBAL LIGATION

Eligibility :
- Age : ≥28yrs & <49yrs.
- married women only.
- At least 1 child, ≥ 1yr of age.

Pre-requisite :
- Consent of female.
- mentally sound individual.
- If mentally unfit patient → Certification from psychiatrist. + Consent of guardian.
- Husband : No h/o vasectomy (Except in case of failure).

Eligibility criteria to perform tubal ligation :
1. MD/DNB/DGO in OBG.
2. MBBS. } Assisted in 5 & performed 5 as a primary surgeon.
3. MS. }

Eligibility criteria to perform vasectomy :
1. Trained MBBS.
2. MS.
3. MD/DNB/DGO in OBG.

Eligibility to undergo vasectomy : < 60yrs.

TUBAL LIGATION (Permanent method)

Preliminary investigations before tubal ligation :
1. Elicit history thoroughly (Obstetric/menstrual/Contraceptive)
2. Physical examination : P/A, P/S, P/V.
3. Blood group.
4. Haemoglobin.
5. Urine routine/microscopy.
6. UPT : If required.
7. Other investigations : As indicated (Eg : USG for bulky uterus).

Patient categorisation :
A : Acceptable.
C : Caution.
D : Delay procedure.
S : Special supervision.

Note :
Tubal Ligation : No C/I.

Indications for delayed procedure :
1. Hb < 7g/dl.
2. H/o APH/PPH (Antepartum/Postpartum haemorrhage).
3. Pregnant.
4. Post partum period : 8-41 days (1-6 wks).
5. Severe pre-eclampsia/eclampsia.
6. Current PID.
7. H/o premature rupture of membrane > 24hrs or h/o puerperal sepsis.
8. Current myocardial infarction.
9. Current DVT/Pulmonary embolism.
10. Cancer : Endometrium/Cervix/GTN (Gestational Trophoblastic Neoplasia).

Tubal ligation safe in :
- BP ≥ 160/100.
- HIV positive patients.
- Diabetics.
- Breast cancer.
- Smokers.

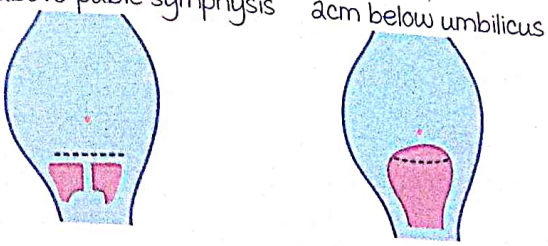
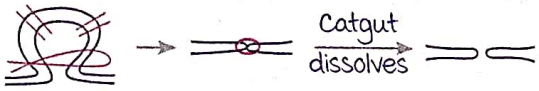
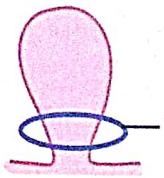
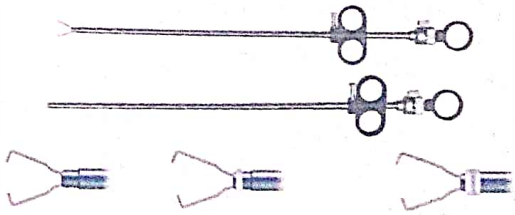
Timing of Tubal Ligation :
- Interval : Done in females not pregnant in the recent past or ≥ 6 wks post-partum.
- Post-partum : ≥ 24hrs of delivery & within 1 week of delivery.
- During caesarian section : Done concurrently.
- Post abortion/Post MTP : Provided medical eligibility criteria fulfilled.

Best time : D1-D7 of cycle.

Routes of Tubal Ligation & Hysteroscopic Tubal Occlusion

00:18:05 ----- Active space -----

ROUTES OF TUBAL LIGATION

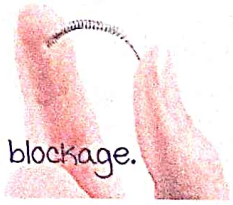
	Minilaparotomy	Laparoscopic Tubal Ligation
Timing	• Post partum. • Interval ligation.	• Only done as interval ligation. • m/c method.
C/Is	-	• Post partum (Risk of uterine perforation) • 2nd trimester abortion.
Techniques	1. Incision Interval ligation → 2cm above pubic symphysis Postpartum → 2cm below umbilicus  2. Loop of isthmus of fallopian tube ↓ Ligated using absorbable sutures Pomeroy : Chromic catgut. Modified Pomeroy : Plain catgut. (Faster absorption : ↓Recanalization)  Catgut → dissolves	Rings In India : Fallope rings. Outside India : Hulka/Filshie clips.  Fallope ring applied at isthmus of fallopian tube using ring applicator  Extended tip (ET) · ET with (1) silastic rings · ET with (2) silastic rings Lap. tubal ring applicator

HYSTEROSCOPIC TUBAL OCCLUSION

Permanent method.

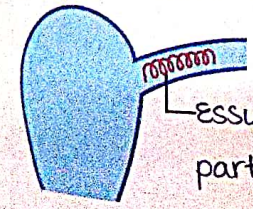
Technique :

Essure : Coiled device made of → Outer : Nickel + Titanium.
→ Inner : Stainless steel.



Essure

tubul blockage.



Essure device introduced into intramural part of fallopian tube hysteroscopically. → Inflammation → Adhesions → Occlusion

Advantage : 99% success rate .

Disadvantage : Requires backup method x 3 months F/b → HSG to confirm complete tubul blockage.

Note : m/c method of → Female sterilization : Laparoscopic tubal ligation.
→ Post partum sterilization : mini Laparotomy.

----- Active space -----

Emergency contraceptives

00:30:18

AKA Interceptives.

Indications :
- After unprotected intercourse.
- Rape victim.
- Condom rupture.

Timing :
- Ideal : within 72 hrs.
- Can be given : upto 120 hrs.

Failure rate : 1.1% → In accidental pregnancy : No risk of
- Teratogenicity.
- Ectopic pregnancy.

Levonorgestrel (1-pill) :
- m/c used.
- Only used upto 72 hrs.
- Mechanism of action :
  1. Delays LH surge → Delays ovulation.
  2. Delays fertilization.

Note : LNG → No effect on implantation. / Not an abortifacient..

Most effective : Cu-IUCD
- Acts immediately
- Prevents implantation.

Hormonal : Ulipristal

→
- Prescription : Not required.
- Can be used multiple times.
- No teratogenicity.

Patient vomits ≤ 1 hr of taking pill : Repeat course.

Drugs used :

1. Levonorgestrel (LNG) - Plan B method :
   - Dose : 1.5 mg single dose or 0.75 mg $\xrightarrow{12\text{hrs later}}$ 0.75 mg.
   - No prescription required.
2. OCPs - Yuzpe method :
   - 100 mcg EE + 0.5mg Progesterone $\xrightarrow{12\text{hrs later}}$ Repeat dose.
   - ↓ Dosing → High dose : 4 pills $\xrightarrow{12\text{hrs later}}$ 4 pills. / Low dose : 2 pills $\xrightarrow{12\text{hrs later}}$ 2 pills.
3. Ulipristal acetate :
   - Dose : 30 mg, OD.
   - more effective than LNG.
   - Indications :
     i. Near mid-cycle.
     ii. ≥72 hrs after unprotected intercourse.
     iii. Obese patients.

----- Active space -----

4. Mifepristone :
   - Progesterone antagonist.
   - Dose : 10mg, oral.
5. Cu-IUCD
6. Centchroman : Maybe tried.

Never used :
1. Misoprostol
2. Mirena
3. Minipill
4. Danazol

Post-partum contraception

00:42:06

Rule of 3 :

Timing of initiation

- Not breastfeeding/ Partially breastfeeding → 3 weeks
- Exclusively breastfeeding → 3 months

Day-1 after delivery :
- Condoms
- Centchroman
- POPs
- Progesterone implants

POST-PARTUM CONTRACEPTION

Other contraceptives :

1. OCPs :
   - ↓Breast milk in breastfeeding females.
   - ↑Risk of thrombosis in breastfeeding & non-breastfeeding females.

Nonbreastfeeding :
- Can start : ≥ 3 weeks.
- Ideally : ≥ 6 weeks.

Exclusively breastfeeding :
- Can start : ≥ 6 weeks
- Ideally : ≥ 6 months.

2. DMPA : Leads to osteoporosis if given ≥ 4wks.

3. IUCD :
   - Postplacental : Within 10 mins of delivery.
   - Postpartum : Till 48 hours of delivery.
   - Interval : After 6 weeks.

4. Emergency contraception : After 4 weeks.

5. Tubectomy :
   - Laparoscopic tubal ligation : C/I in postpartum period.
   - Minilaparotomy (modified Pomeroy technique) :
     - Postpartum : Till 1 week.
     - Interval : After 6 weeks.

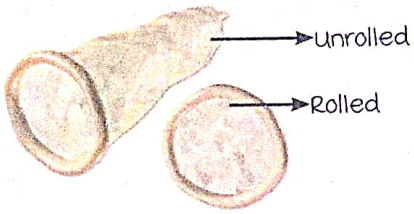
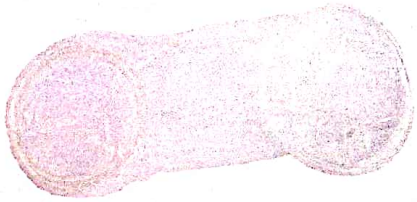
----- Active space -----

CONTRACEPTION : IMAGES

Barrier methods

00:00:26

Condoms :

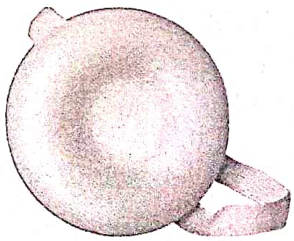
	Male Condoms	Female Condoms
Ring	 Single ring	 2 rings
Appearance	A pointed end.	Short, broad & no pointed end.
Made of	Polyurethane condom.	• Female condoms 1 : Polyurethane. • Female condoms 2 : Nitrile.
Coitus	Coitus dependent : Used on erect penis.	Coitus independent.
Failure rate	2-15/HWY.	5-20/HWY.
Programme	Included in National Family Planning Programme : as Nirodh.	Distributed by NACO to female sex workers : Free of cost.
Usage	Single use.	Can be used twice.

HWY : Hundred women years.

Advantage : Protects against HIV (male condom > Female condom), PID, STD, cervical dysplasia, pregnancy.

Today Sponge :

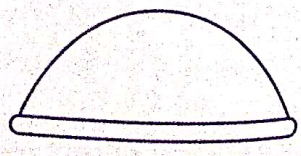
- Mushroom shaped device.
- 1 gm of nonoxynol-9 (Spermicide).
- Used multiple times for 24 hrs after insertion.
  - Max permitted duration : 30 hrs.
  - Chances of toxic shock syndrome : Nil.



Today Sponge

Diaphragm :

- Dome shaped device with flexible rim.
- Washed and reused upto 1 year.
- Made of rubber/latex/silicone.
- Not sperm-proof.
- Spermicidal agent is added.



Diaphragm

00:07:37 ----- Active space -----

Mala N & D :

- Composition : Ethinylestradiol 30 mcg + Levonorgestrel (LNG) 150 mcg
- 21 hormonal pills + 7 ferrous fumarate tablets.

Mala N : Distributed by government of India free of cost.

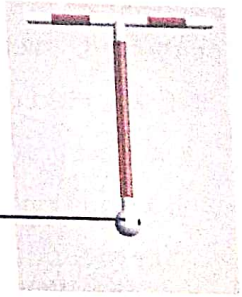
Mala D : Distributed at subsidized rates.

IUCDs :

- 1st gen : Lippes loop.
- 2nd gen : Copper IUCD.
- 3rd gen : Hormonal IUCD (mirena).
- 4th gen : Frameless IUCD.

Copper T 380 A (Paraguard) :

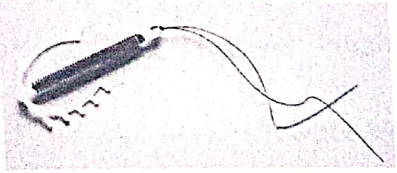
- T shaped device.
- 380 mm^2 : Surface area of copper wire.
- A : Copper present on the arms of T tube.
- 50 mcg of copper released per day.
- Life span : 10 yrs.



Copper T 380 A

Multiload 375 :

- Copper present on verticle stems.
- Bent arms with spurs → Spurs lessen expulsion.
- Thread made of nylon.
- Life span : 5 years.


Multiload 375

Mirena (LNG 20) :

- Contains LNG.
- Used in AUB, Endometrial hyperplasia without atypia.
- Not used as emergency contraception.
- Releases 20 mcg/day
- Life span : 5 yrs.
- MOA : Prevents implantation (Does not cause anovulation).

mirena

----- Active space -----

Side effect :

- Irregular bleeding.
- Endometrial atrophy on continuous use → Amenorrhoea.

Gynefix IUCD :

- Frameless IUCD : 4th gen IUCD.
- Aka Copper fix IUCD/Flexiguard.
- Used in nulliparous females.
- Copper cylinders with nylon thread.
- Sutured to myometrium.

Advantages :

- Less pain.
- Less expulsion rate.

Gynefix

Implants

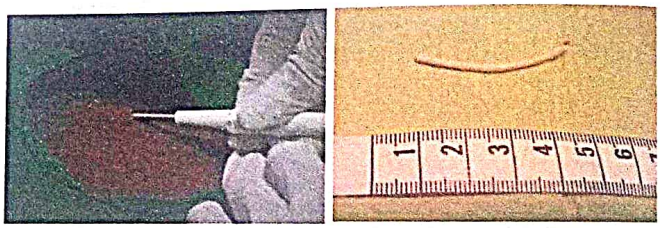
00:17:21

Implanon :

- Single rod implant.
- Contains etonogestrel 68 mg → Released at 60 mcg/day.
- Inserted on non-dominant arm (medial side) on D_1-D_7 menstrual cycle.
- If inserted after D_7 : Added contraception required x 3 days.
- Long acting reversible contraceptive (LARC).
- Not provided free of cost by GOI.

Nexplanon :

- Barium sulfate in stem.
- Not radiopaque.



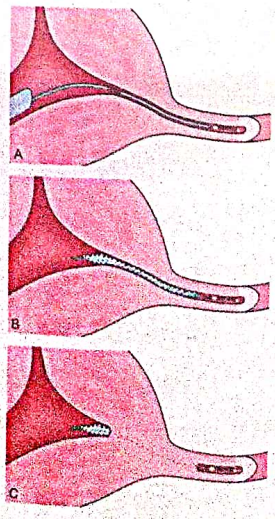
Implanon

Note :

LARC : Implants, IUCD, Injectables.

Essure :

- Coiled device with outer & inner cylinder.
  - Outer cylinder : Nickel + Titanium.
  - Inner cylinder : Stainless steel.
- Inserted into Intramural/Interstitial part of fallopian tube via hysteroscopy.
- Tubal occlusion by inflammation :
  - Occlusion by 3 months.
  - Confirmation of occlusion : HSG after 3 months.
  - Female advised for other contraceptives until 3 months.
- Success rate : 99%.

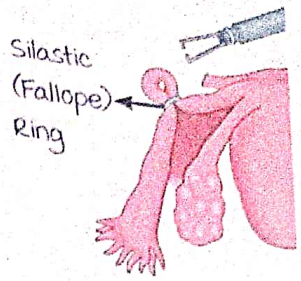


Essure

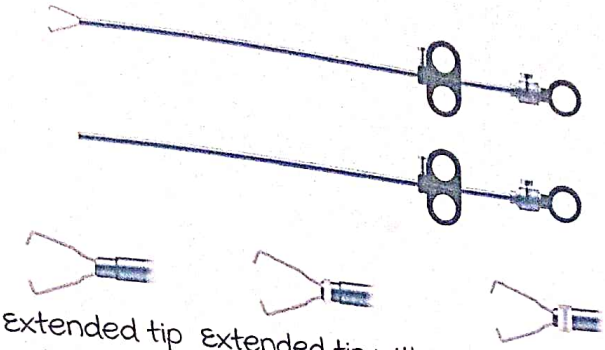
00:21:46 ----- Active space -----

LAPROSCOPIC TUBAL LIGATION

- m/c method of ligation.
- Ring : Fallope rings.
- Rings applied in isthmus part of fallopian tube.
- Not used in :
  a. Post-partum period.
  b. 2nd trimester abortion.



Lap. Tubal Ligation

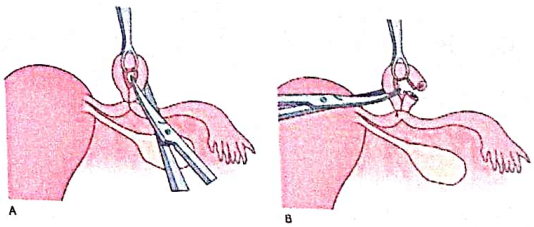


Extended tip | Extended tip with (1) silastic rings | Extended tip with (2) silastic rings

Lap. tubal ring applicator

MINI LAP TECHNIQUE

1. Loop made in isthmus of fallopian tube.
2. Loop occluded with sinle/double ligature.
3. Loop is cut.



Mini Lap. Technique

mlc techniques

Pomeroy :

- Chromic catgut
- Absorbable
- Takes time to dissolve.

Modified pomeroy :

- Plain catgut
- Absorbable
- Dissolves quickly.

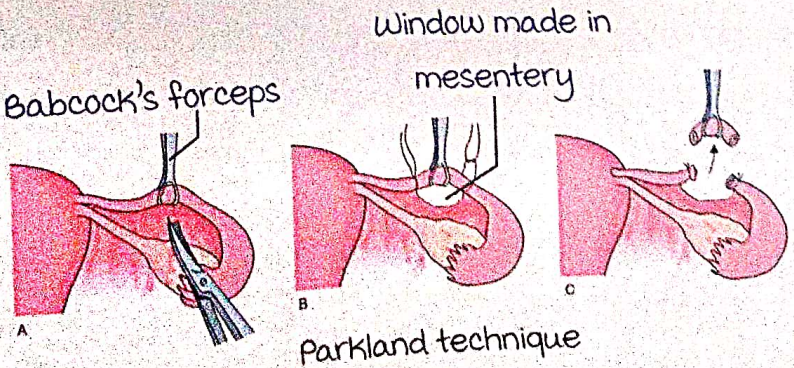
Done :

1. Postpartum.
2. As interval sterilization.

Parkland technique :

1. Double ligation made in the loop of isthmus.
2. Loop is cut.

Advantages : Immediate separation of two ends of tube.



Parkland technique